1)

int x = 5; → 1
for(int i = 1; i <= n; i++) → End-Start +1 +1 = n -1+1+1=n+1
{
for(int j = 1; j <= n; j++) → n-1+1+1=n+1
{
x = x + i + j; → n
s.o.p(x); → n
}
}

× n

SE = $1+n+1+n^2+n+ n^2+ n^2=3n^2+2n+2$ Big O = $O(n^2)$

---

2)

int a = 0, i = n; →2
while (i > 0) → $\log_2 n$
{
a += i; → $\log_2 n-1$
i /= 2; → $\log_2 n-1$
}

SE = $2+\log_2 n+ \log_2 n-1+ \log_2 n-1= 3\log_2 n$ Big O = O(log n)

---

3)

int x=0; →1
for(int i=0; i<=n ; i++) → End-Start +1 +1 = n -0+1+1=n+2
for(int j=3 ; j <n ; j++) → n -3+1=n-1
{
System.out.println(j); →n-2
x=x+j; → n-2
}

SE =1+n+2+n-1+n-2+n-2=4n-2 Big O = O(n)

4)

int a = 0, b = 0; $\rightarrow$2

for (i = 0; i < N; i++) $\rightarrow$n-0+1 =n+1

{

a = a + rand(); $\rightarrow$n

}

for (j = 0; j < M; j++) $\rightarrow$m-0+1=m+1

{

b = b + rand(); $\rightarrow$m

}

SE =2+n+1+n+m+1+m=2n+2m+4 Big O = O(n)

---

5)

int s=0; $\rightarrow$1

for(int i =1; i<=n; i=i+4) $\rightarrow \frac{n-1}{4}+1+1=\frac{n-1}{4}+2$

{

s=s+1; $\rightarrow \frac{n-1}{4}+1$

S.O.P(s); $\rightarrow \frac{n-1}{4}+1$

}

SE $=1+\frac{n-1}{4}+2+\frac{n-1}{4}+1+\frac{n-1}{4}+1=\frac{3(n-1)}{4}+4$ Big O = O(n)

---

6)

int x = 5; $\rightarrow$1

for(int i = 1; i <= n; i++) $\rightarrow$n-1+1+1=n+1

{

for(int j = 1; j <= i; j++) $\rightarrow \frac{n(n+1)}{2}+1$

{

x = x + i + j; $\rightarrow \frac{n(n+1)}{2}$

s.o.p(x); $\rightarrow \frac{n(n+1)}{2}$

}

}

SE $=1+n+1+\frac{n(n+1)}{2}+1+\frac{n(n+1)}{2}+\frac{n(n+1)}{2}=\frac{3n(n+1)}{2}+n+2$ Big O = O(n)

7)

```
static int power(int a, int b)
{
if (b < 0) return a; →1
if (b == 0) return 1; →2
int sum = a; →1
for (int I = 0; I < b - 1; I++) →(b-1)-0+1=b
{
sum *= a; →b-1
}
return sum; →1
}
```

SE =1+2+1+b+b-1+1=2b+4 Big O = O(n)

---

8)

```
int i = 0; →1
while (i < 5) →5-0+1=6
{
System.out.println(i); →5
i++; →5
}
```

SE =1+6+5+5=17 Big O = O(1)

---

9)

```
x = 1; →1
for (i = -3; i < n; i ++) →n+3+1=n+4
{
x = x + i; → n+3
y = x + 4; → n+3
}
```

SE =1+n+4+n+3+n+3=3n+11 Big O = O(n)

10) Compute S.E and big O for the following code then substitute n by 4 :

```
for(int i = 1; i <= n; i++) →n-1+1+1=n+1
{
s.o.p(x); →n
for(int j = 1; j < n; j++)→n-1+1=n
{                                      × n
x = x + i + j; →n-1
}
}
```

SE $=n+1+n+n^2+ n^2-n=2n^2+n+1$ Big O = $O(n^2)$

substitute n by 4 $=2(4)^2+4+1=37$

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Class : 6